

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Celebal Technologies Internship

Submitted By

Assignment No-3: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Dataset: Sample Superstore Dataset

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Technology: Microsoft Azure & Azure Data Factory

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1. Introduction

Cloud computing enables organizations to store, process, and analyze data without maintaining physical servers. Microsoft Azure provides various cloud services that help organizations build secure and scalable data solutions.

In this assignment, an end-to-end data pipeline was developed using Azure Blob Storage and Azure Data Factory (ADF). The pipeline reads a CSV file from a source Blob container, validates the file using the Get Metadata activity, copies the file to another Blob container, and monitors the execution

2. Objective

The objective of this assignment is to understand Azure cloud fundamentals and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory.

The objectives achieved include:

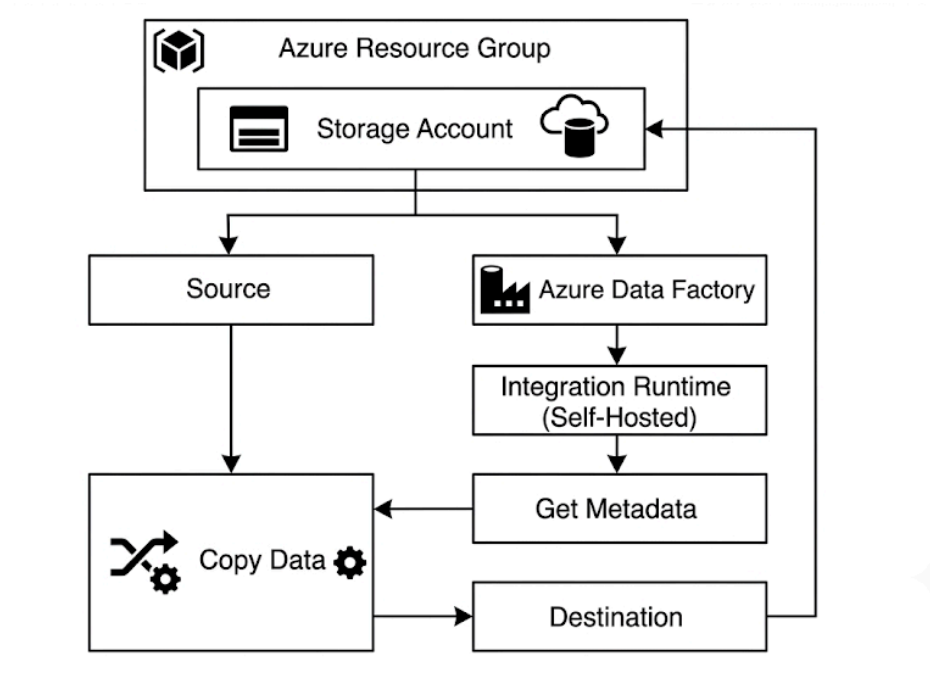
- Understanding Azure Portal
- Creating Azure Resource Group
- Creating Azure Storage Account
- Uploading CSV data into Blob Storage
- Creating Azure Data Factory
- Creating Linked Services
- Creating Source and Destination Datasets
- Using Get Metadata Activity
- Using Copy Data Activity
- Running and Monitoring Pipelines
- Implementing IAM Roles
- Building an end-to-end Blob → ADF → Blob pipeline

3. Azure Services Used

Service	Purpose
Resource Group	Logical container for Azure resources
Storage Account	Stores Blob containers and CSV files
Blob Storage	Stores source and destination files

Azure Data Factory	Creates and executes data pipelines
IAM	Provides secure access using Azure RBAC

4. Architecture Diagram



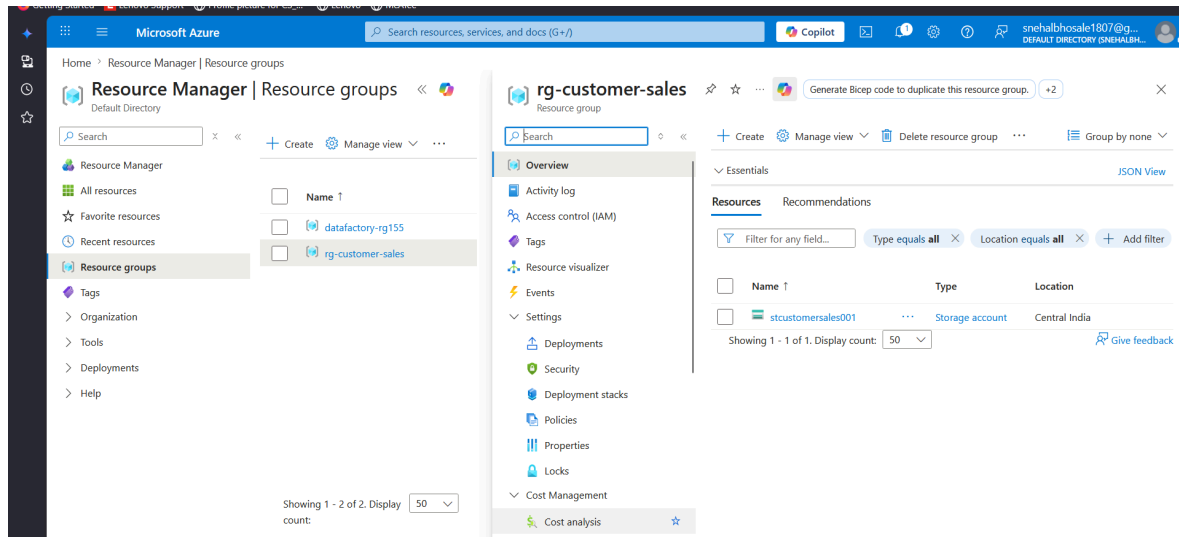
Task 1 – Resource Group Creation

Description

A Resource Group named **rg-customer-sales** was created to organize all Azure resources used in this project.

The Resource Group acts as a logical container that simplifies resource management, deployment, and deletion.

Screenshot



Task 2 – Storage Account Setup

Description

A Storage Account named **stcustomersales001** was created inside the Resource Group.

Two Blob Containers were created:

- Source
- Destination

The Sample Superstore CSV dataset was uploaded into the Source container.

Source Container

- Container Name: source
- Uploaded File:

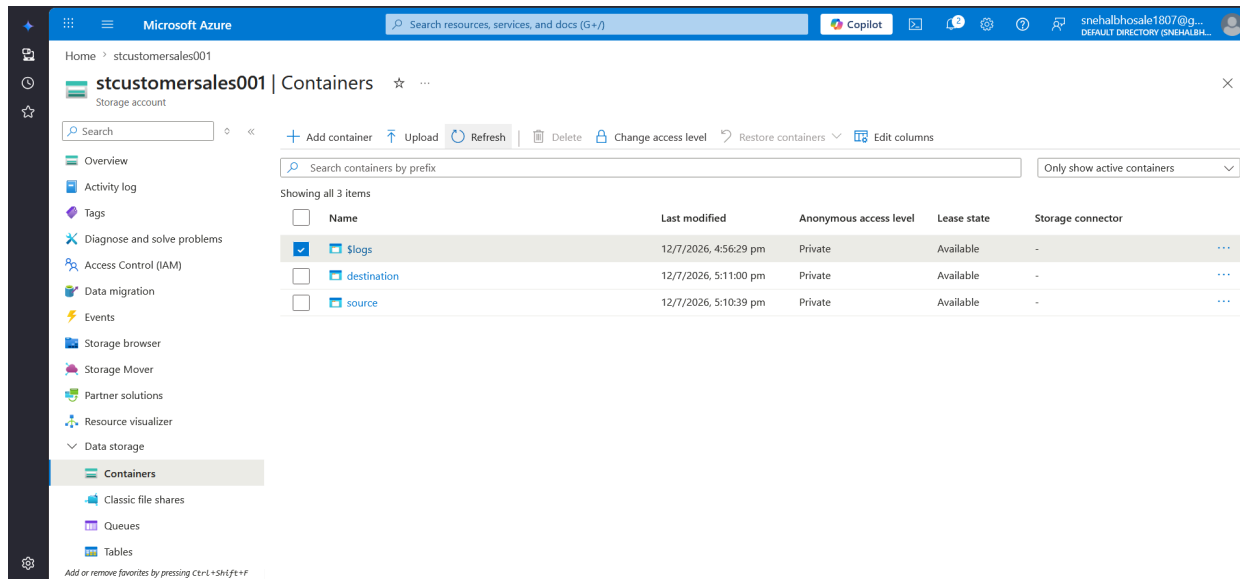
Sample - Superstore.csv

Destination Container

- Container Name:

destination

Initially, the destination container was empty because the output file would be generated after pipeline execution.



Task 3 – Azure Data Factory Configuration

Azure Data Factory

An Azure Data Factory instance named
adf-customer-sales
was created.

ADF Studio provides three important sections:

Author

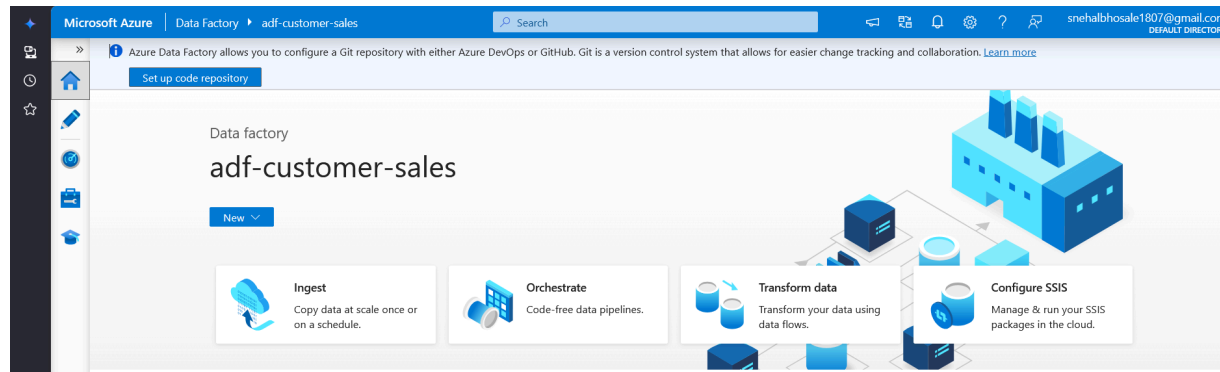
Used to create pipelines, datasets and linked services.

Monitor

Used to monitor pipeline execution.

Manage

Used to create Linked Services and Integration Runtime.



Linked Service

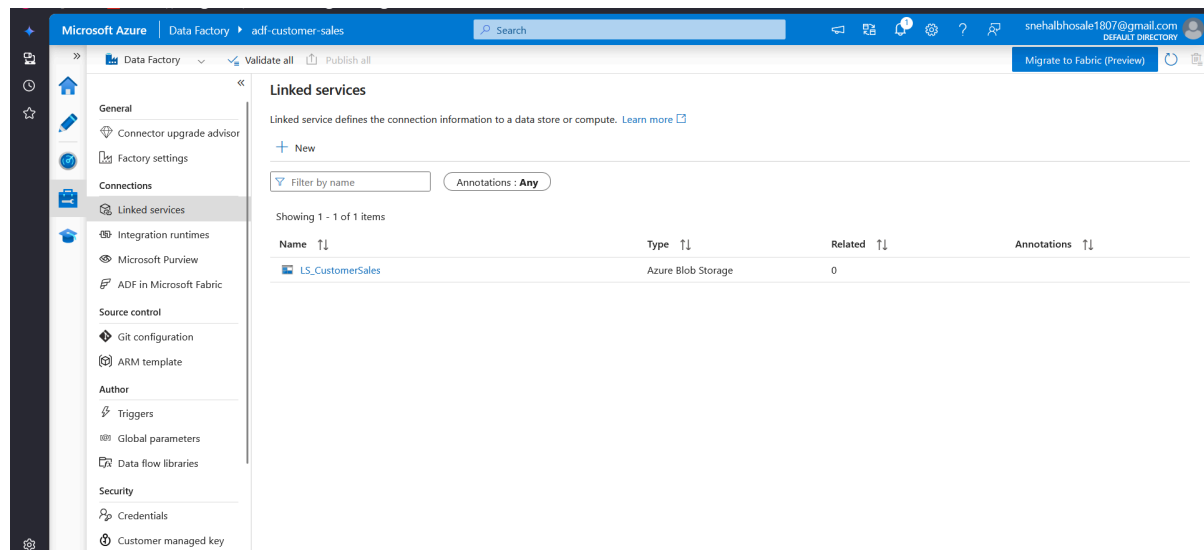
A Linked Service was created to connect Azure Data Factory with Azure Blob Storage.

Linked Service Name

LS_CustomerSales

Authentication

Account Key



Source Dataset

Dataset Name

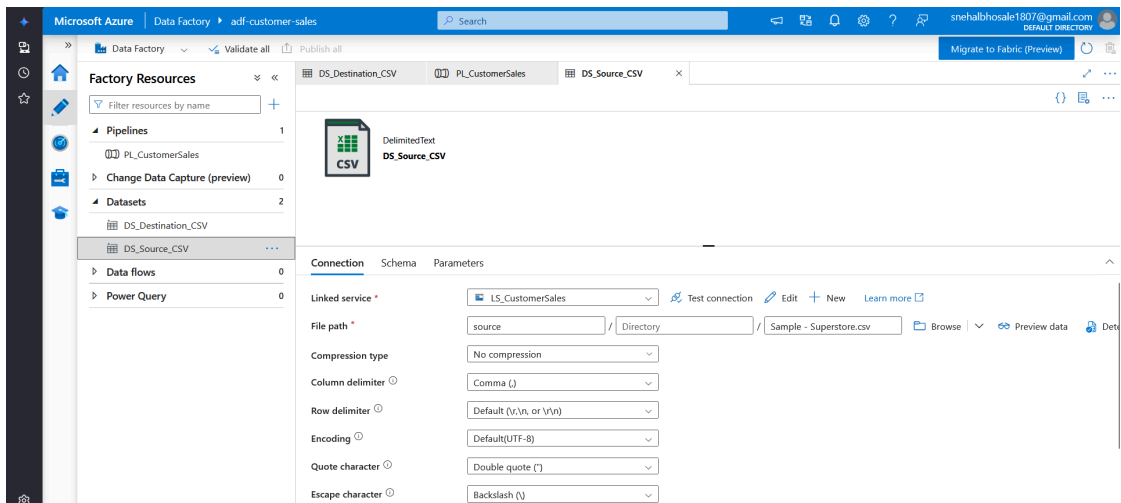
DS_Source_CSV

Container

source

File

Sample - Superstore.csv



Preview data

Linked service: LS_CustomerSales

Object: Sample - Superstore.csv

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
1	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
2	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
3	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal
4	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CM4500 Series Slim Rectangular Table
																	Eldon Fold

Destination Dataset

Dataset Name

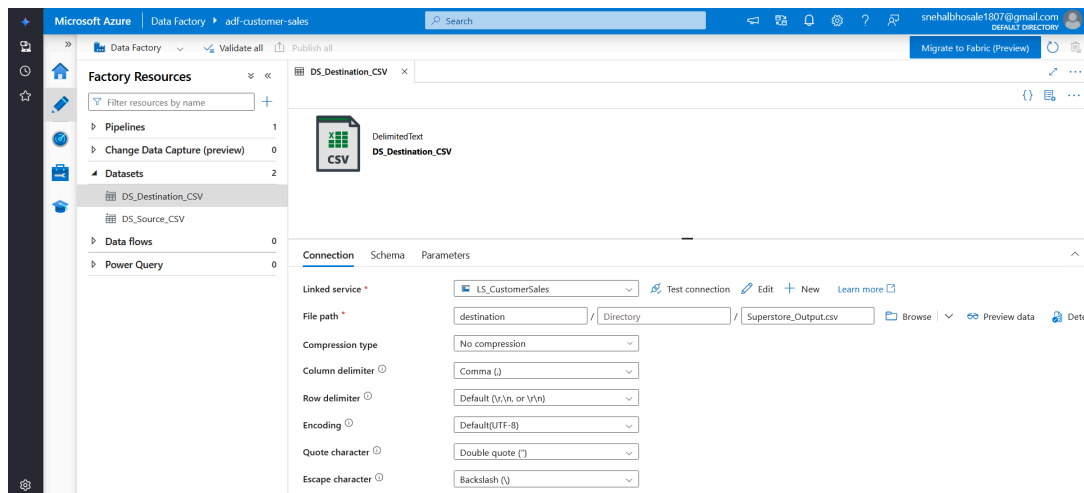
DS_Destination_CSV

Container

destination

Output File

Superstore_Output.csv



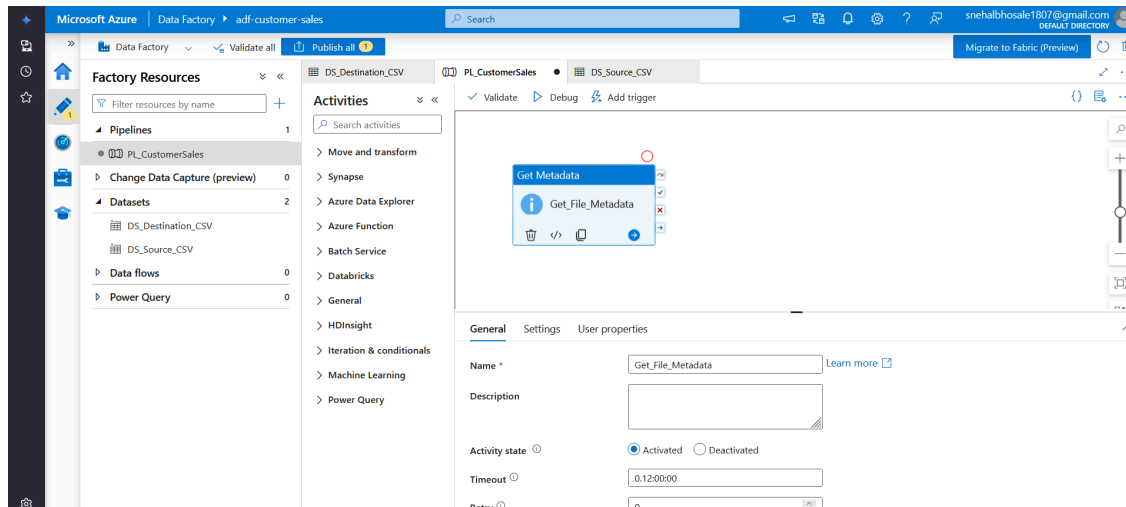
Get Metadata Activity

Purpose:

Checks whether the source file exists and retrieves metadata before processing.

Metadata Retrieved:

- Exists
- File Size
- Last Modified Date



Task 4 – Pipeline Development

Pipeline Name

PL_CustomerSales

The pipeline consists of two activities.

Copy Data Activity

Purpose:

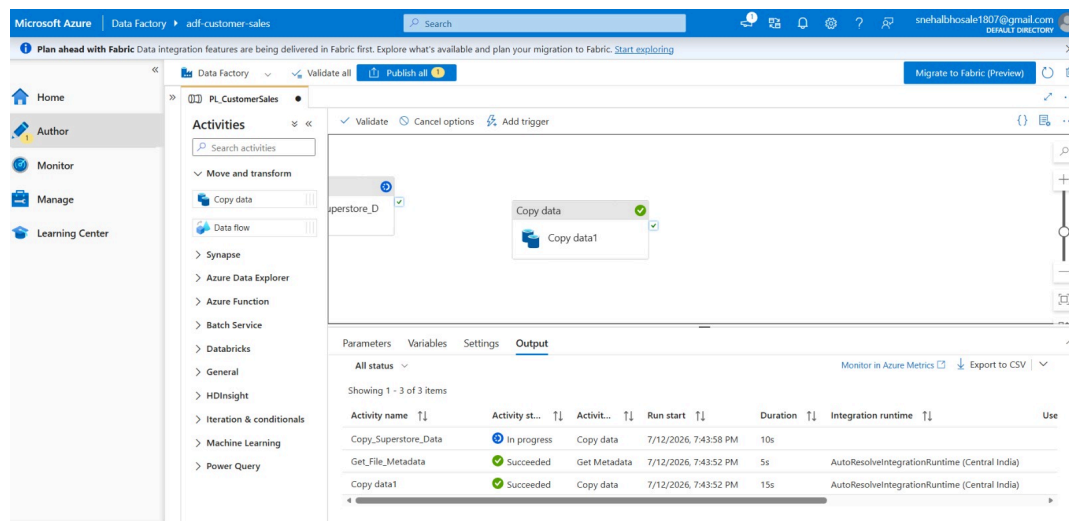
Copies the CSV file from the Source container to the Destination container.

Source

DS_Source_CSV

Destination

DS_Destination_CSV



Pipeline Workflow

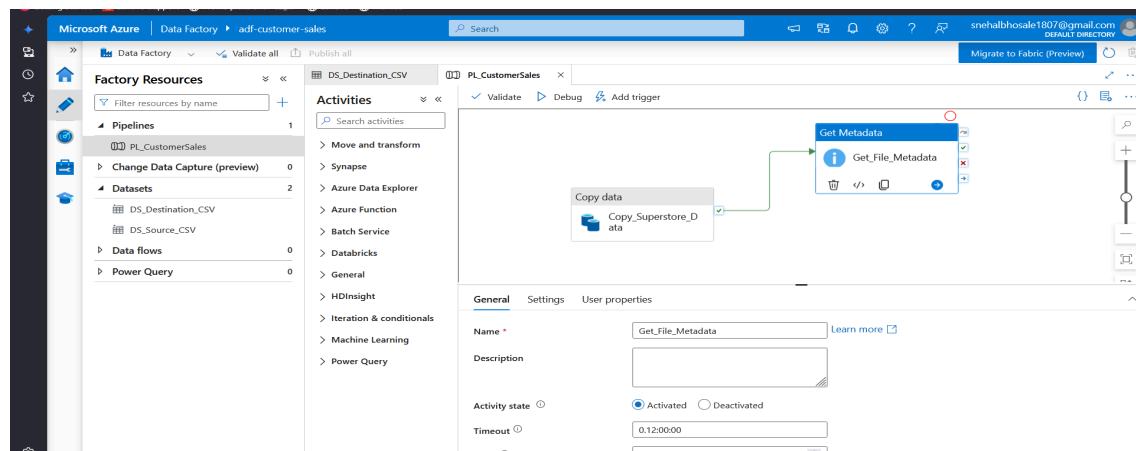
Get Metadata



Copy Data



Destination Blob



Task 5 – Pipeline Execution

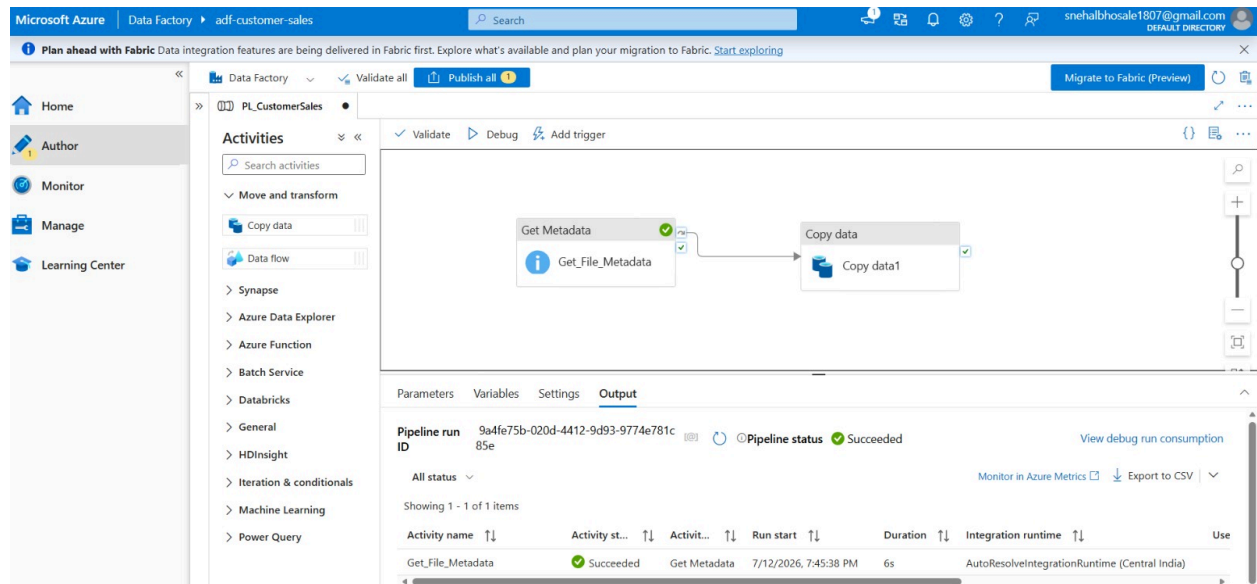
The pipeline was validated successfully before execution.

Validation ensured that there were no configuration errors.

The pipeline was executed using the Debug option.

After successful execution, the Monitor section displayed the pipeline status as **Succeeded**.

The output file **Superstore_Output.csv** was successfully created in the Destination Blob container.



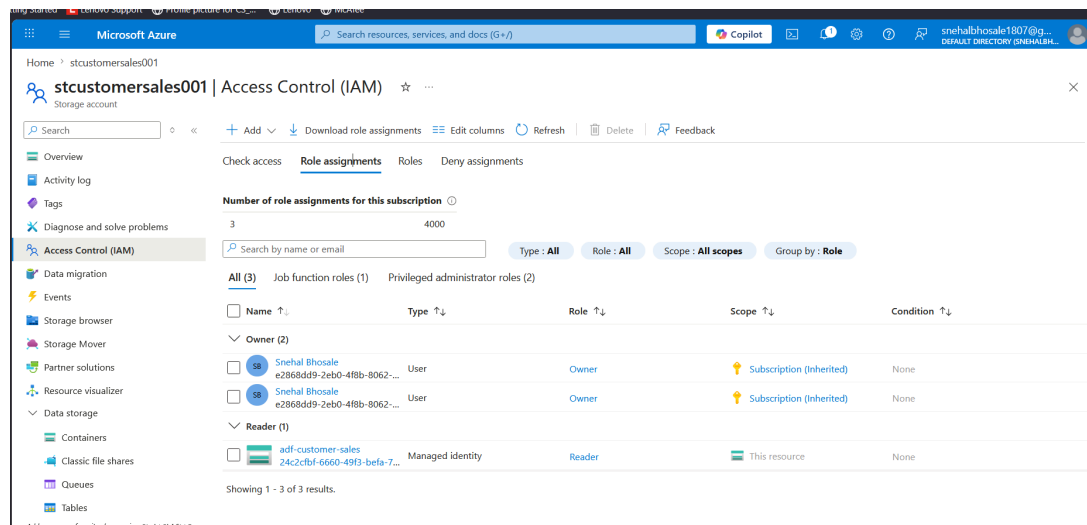
Task 6 – IAM Role Assignment

Azure Role-Based Access Control (RBAC) was used to provide secure access between Azure Data Factory and Azure Storage.

The following roles were assigned:

- Reader
- Contributor

These permissions allow Azure Data Factory to access the Storage Account and perform data copy operations securely.



Learning Outcomes

Through this assignment, I gained practical knowledge of Microsoft Azure and its core cloud services. I learned how to create and manage a Resource Group, Storage Account, and Blob Containers for storing data in the cloud. I also understood how to create an Azure Data Factory instance, configure Linked Services and Datasets, and build an end-to-end data pipeline.

Additionally, I learned to use the **Get Metadata** activity to validate source files before processing and the **Copy Data** activity to transfer data from the source container to the destination container. I also gained experience in validating, publishing, executing, and monitoring pipelines, as well as understanding the importance of Azure IAM roles in providing secure access between Azure services.

Overall, this assignment enhanced my understanding of cloud-based data integration and provided hands-on experience with Azure Data Factory, which is an essential tool for modern data engineering.

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end data pipeline using Microsoft Azure and Azure Data Factory. The required Azure resources were created, including a Resource Group, Storage Account, Blob Containers, and Azure Data Factory. A Linked Service and Datasets were configured to establish connectivity between Azure Blob Storage and Azure Data Factory.

The pipeline was designed using **Get Metadata** and **Copy Data** activities to validate the source file and copy the CSV file from the source container to the destination container. The pipeline

was executed successfully, and the output file was generated in the destination location. The execution was monitored through Azure Data Factory, and appropriate IAM roles were configured to ensure secure access.

This assignment provided valuable hands-on experience in designing, implementing, and managing cloud-based data pipelines, strengthening my understanding of Azure cloud services and data engineering concepts.